

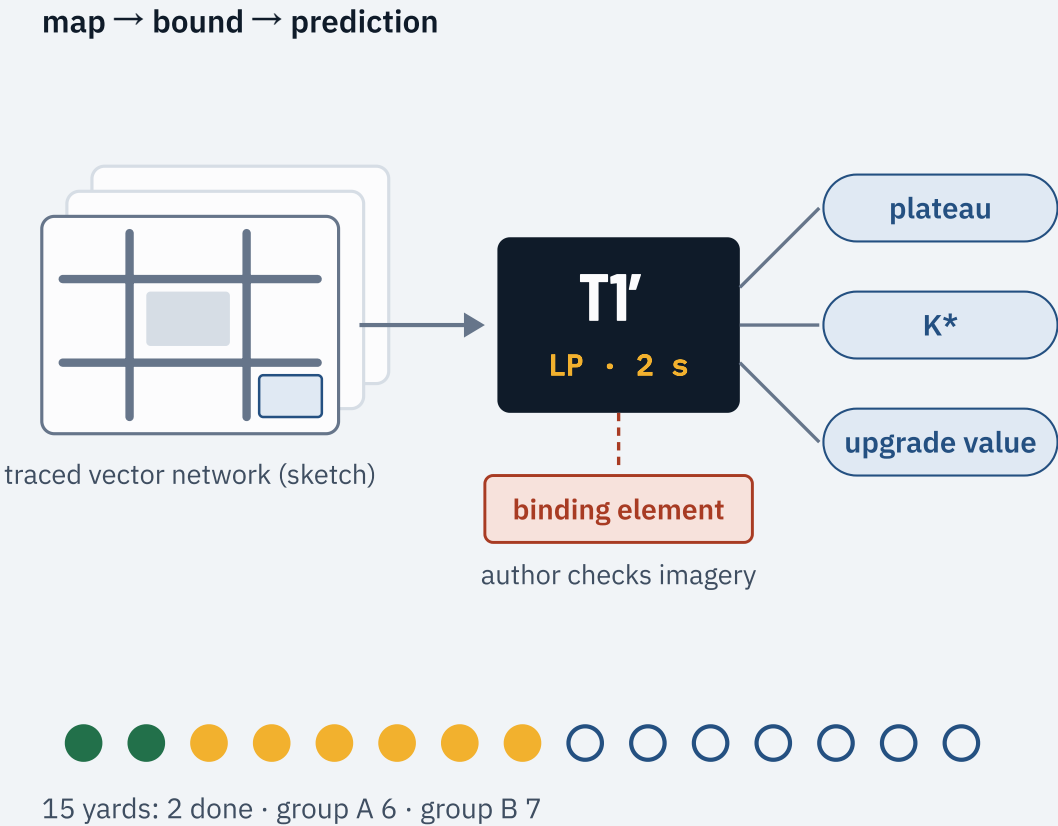
One map, a two-second LP: can it predict a new shipyard's capacity?

Testing across shipyards: the network bound computed from the map, with its binding elements, as a cross-yard predictor (C5, H6)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts (working title)

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2 yards done · group A's 6 being annotated · 15 in all



This sub-study in one minute

1

The question

Do capacity results from two yards carry over? Chapter 6 tests C5: the bound computed from the map, with its binding elements, predicts plateaus, jamming thresholds and upgrade values across yards.

2

The test

The predictor is T1' itself, a two-second LP per yard, with its binding elements; tested by leave-one-out, yards grouped by key element. Topological indicators explain where the bound binds; they do not replace it.

3

Two points so far

Okpo is a dock-road yard (road 161), Yantai a door-type yard (building 014). On both, the reservation plateau is about 36% of T1'; each predicts the other within 0.5%. Yantai's daily tasks are 2–4% of T1'.

4

Next

The author is annotating group A's six yards. Each gets T1' first (fast), then a check of its top three key elements on imagery and the granularity, then simulations. Group B follows at eight yards; 15 in all.

Bound first, binding element next, simulation last: screening a new yard starts with a two-second LP.

The question: which yards do the earlier results carry over to?

C5	Transferable The network bound computed directly from the map, and its binding elements, predict plateaus, jamming thresholds and upgrade values across shipyards
H6 Cross-yard prediction	
Hypothesis	The network bound and binding elements computed directly from the map predict the simulated plateau, K^* and upgrade values across shipyards. Topological indicators serve to explain why the bound binds where it does.
Test	Leave-one-out test; classification by key element: dock-road single-lane type, door type, ...
If it holds	A two-second linear program is enough to screen a new shipyard.
If it fails	What transfers is the bound tool and the key-element classification themselves.

Where chapter 6 sits in the thesis

Ch. 3 Capacity	$T1'$, the capacity interval [best deadlock-free rule, $T1'$] and the K^* criterion: the predictor and the first two targets
Ch. 4 Orchestration	Pushes the interval's lower edge up: plateaus move with the rule, so each rule is read separately
Ch. 5 Co-design	Dual prices and upgrade simulations: the third target, upgrade value
Ch. 6 Many yards	Which yards these results carry over to: prediction errors, the key-element classification and a screening procedure

H6 pays either way: if it holds we gain a screening tool for new yards; if not, the bound tool and the key-element classification are themselves what transfers.

The predictor is T1' itself, not a topology regression on 15 samples

Regressing topological indicators on 15 yards

- Fifteen samples support one or two predictors at most and overfit easily
- Traffic rules and task flows sit between indicators and capacity; no direct mechanism

T1' as the predictor

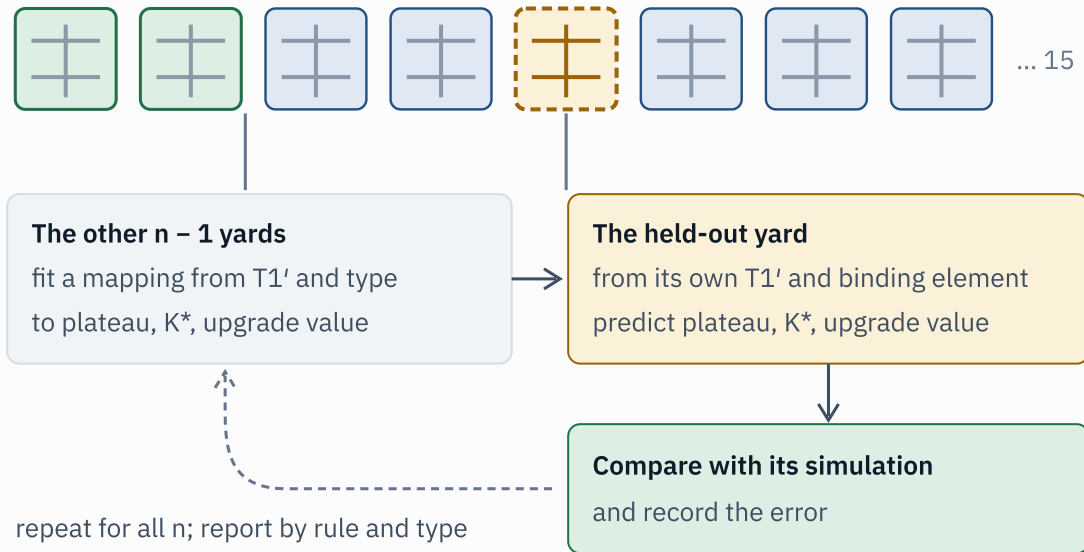
- From the map in two seconds per yard: an occupancy LP with width bin-packing, crane and window rows
- The upper edge of the capacity interval (C1), valid for every rule meeting (A1)–(A3), loaded routes fixed
- Dual prices name the binding element: where to look on the imagery first
- Transfer source: railway capacity without a fixed timetable (Jensen et al. 2020, abstract level)

Topology of the two yards: to explain, not to replace

	OKPO	YANTAI
Roads / centreline	721 / 40.3 km	140 / 8.6 km
Junctions / entrances / stops	334 / 218 / 421	58 / 32 / 140
Independent cycles μ / meshedness α	82 / 0.064	14 / 0.056
Dock-bound minimum cut (sum of widths)	62 m	137 m
Dock entry degree (max)	5	5
→ T1' (a day)	1,720	1,252
→ binding element	dock-road single-lane	door type
→ K*(20%)	50	25
→ reservation plateau (a day)	627	455

Similar loop density, yet K* differs twofold and the binding element differs in type. The indicators explain why it binds there: a narrow dock-bound cut at Okpo, a door at Yantai.

The test: leave-one-out, with yards grouped by key element



Two OD samplings (by entity, by stop) are each yard's standard sensitivity (T19). New types come from the new yards' dual prices, not fixed in advance; each type records the resource granularity it was read under.

Three targets

Plateau

The interval's lower edge, read per traffic rule (chapters 3, 4)

K^*

Network takes over: marginal gain first below 20% of free flow (H1, chapter 3)

Upgrade value

Dual-price prediction against upgrade simulations (chapter 5, from T25)

Two key-element types so far

Dock-road single-lane · Okpo

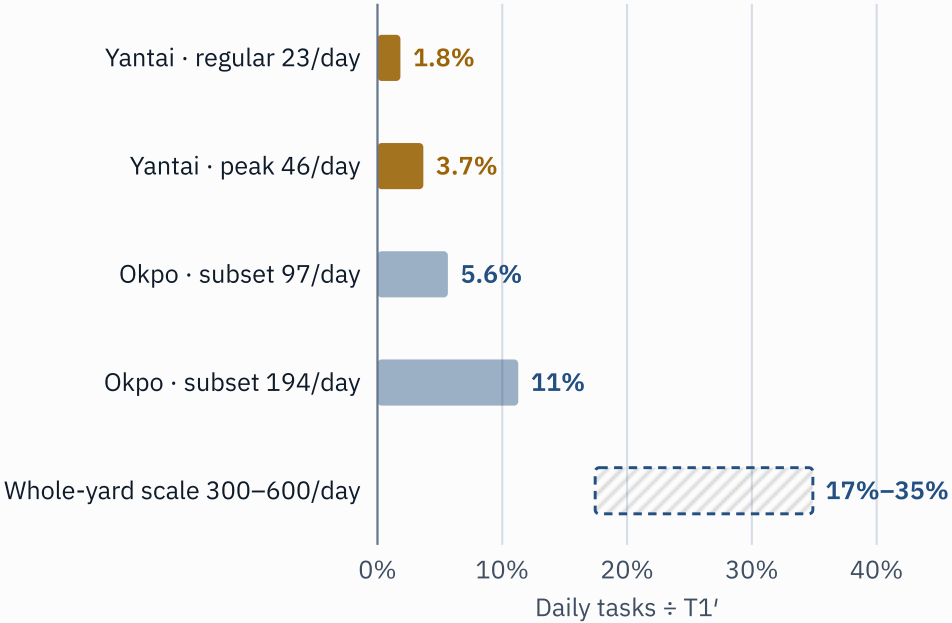
Road 161, dock 1's only stopping road (9 m × 1,095 m); T1' binds there in 5 of 10 seeds, on the narrow road 010-6 in 4.

Door type · Yantai

Building 014's two doors (paint shop, presumed): each takes about 7% of all tasks and is held about 10 min per load; T1' binds there in 9 of 10 seeds.

The first two data points: Okpo and Yantai fall into two key-element types

	OKPO	YANTAI
Map version (same network as the previous one)	derived map rev 1825	r396
Network: roads / centreline	721 / 40.3 km	140 / 8.6 km
T1' (a day; 10 seeds [min, max])	1,720 [1,436, 1,948]	1,252 [1,157, 1,318]
Capacity interval [reservation plateau, T1']	[627, 1,720]	[455, 1,252]
Reservation plateau ÷ T1'	36%	36%
K*(20%) (range)	50 (50–60)	25 (25–25)
Where T1' binds (seeds)	road 161 ×5 · road 010-6 ×4 · road 026 ×1	building 014's two doors ×9 · a door of building 007 ×1
Type	dock-road single-lane	door type
Daily tasks ÷ T1'	4-transporter subset 5.6%–11%; whole-yard caliber awaits T22	1.8%–3.7% (regular–peak)

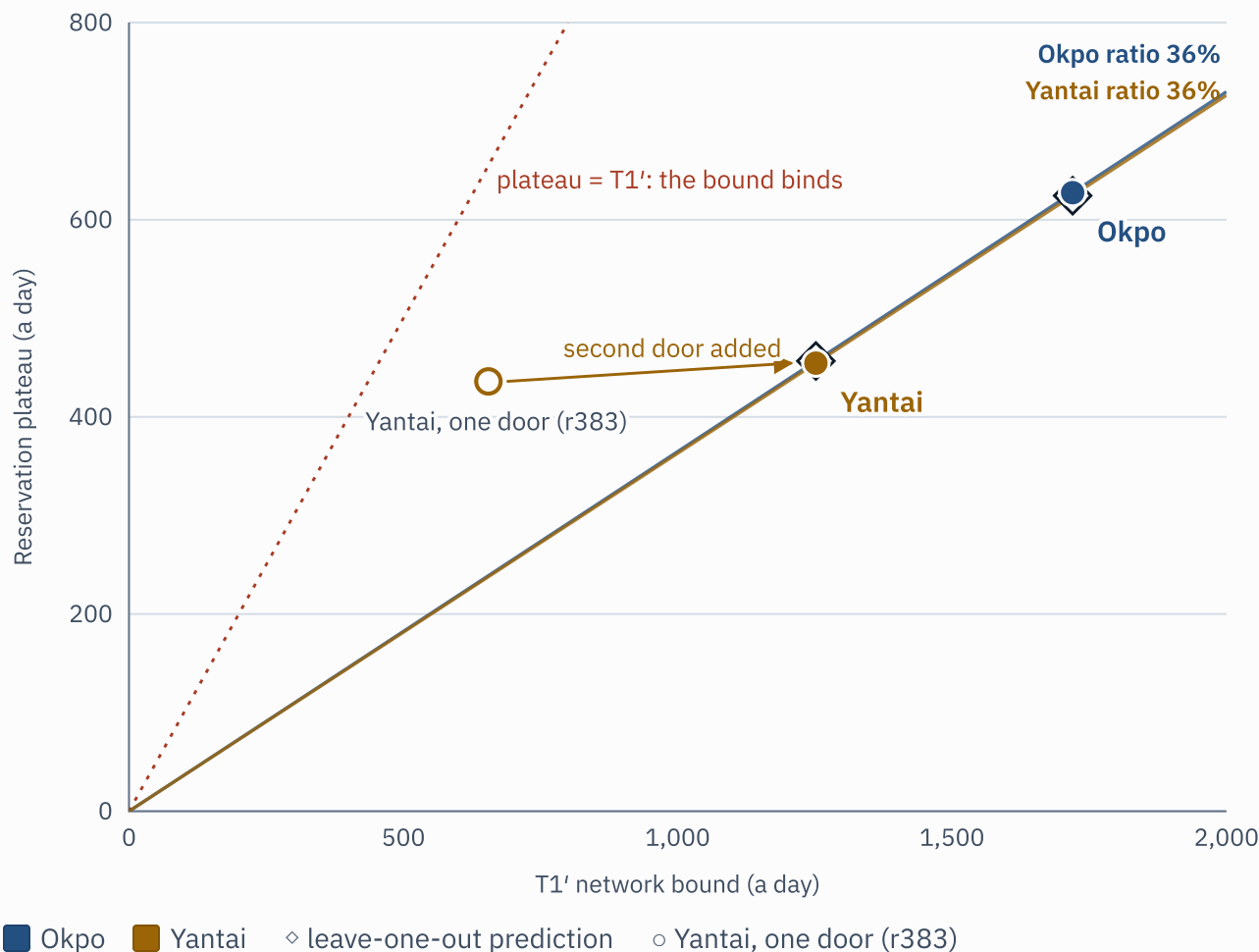


■ Yantai ■ Okpo · 4-transporter subset (Yim 2008), comparison only
▨ public scale of large Korean yards, not Okpo itself; Okpo awaits T22

Density criterion: daily tasks ÷ network bound (H1). One folding method for every yard: Yantai's comes from its public steel use, 68,512 t ÷ 100 t × 10 moves ÷ 300 days ≈ 23 a day.

Sources: thesis-core 83c16bf (T18–T21, each reviewed by the project control and an independent reviewer): bounds.csv of Okpo results/a0ab16fceb94 and Yantai results/b82aa19b22af, t18, t19, t20_summary.json, t20_demand.csv (100 t blocks after Zhang 2025, 10 moves after Wang 2017). Conditions: whole-route reservation (the best deadlock-free rule so far), saturated, MY-B flow mix, plateau = mean over K = 100–150, whole-segment resources (Okpo road 161 whole, Yantai doors held while loading), 10 seeds. Okpo's 97 / 194 are a 4-transporter subset (Yim 2008; Roh & Cha 2011); public whole-yard scale 300–600 a day (Yim 2008; Shen et al. 2018); Okpo's own volume awaits T22 and T24.

First leave-one-out reading: both plateaus are about 36% of T1'



36% / 36%

Reservation plateau ÷ T1': Okpo / Yantai

±0.5%

Error when one yard's ratio predicts the other's plateau (two-point leave-one-out)

67% → 36%

Yantai before and after building 014's second door: T1' 655 → 1,252, plateau 436 → 455

Two points only show how the test runs. The ratio moves with the traffic rule and the OD sampling (35% for Yantai sampled by stop in T19), so chapter 6 reads it per rule and per type. Near 1 the bound binds and road changes pay; far below 1, orchestration comes first: the boundary between C3 and C4.

Sources: thesis-core 83c16bf: T1' = mean of 10 seeds in bounds.csv; reservation plateaus from t18_summary.json and t20_summary.json (mean over K = 100–150); Yantai's one-door setting from results/t20_report_c12289dfa9d/t20_r383_r395.csv. Leave-one-out: one yard's plateau ÷ T1' times the other yard's T1'. Conditions as on the previous page; the C3–C4 boundary is part of C4 in research manifesto v1, §2.

Key-element review: the top three elements of T1' , checked on imagery

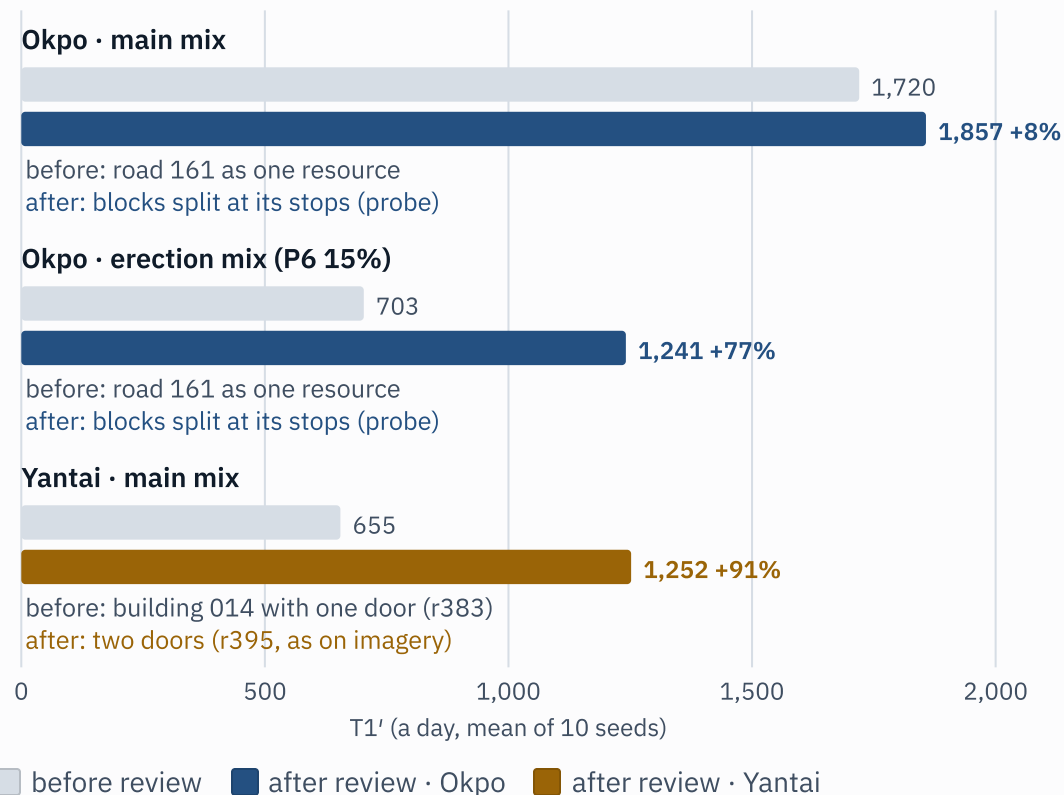
- 1 List the top three elements by dual price**, with building, entrance name and coordinates, not just a label
- 2 The author checks three things on the imagery**: road width; the number of doors; how traffic passes roadside stops and doors (while a vehicle works there, can others get in and out, can they pass?)
- 3 Record it in the map's source field**
- 4 Choose the resource granularity and recompute T1'**: whole road or blocks, door held or shared

Okpo · road 161

The model held the 1.1 km dock road as one resource. Imagery: neighbouring assembly sites stay reachable while a vehicle works at dock 1. → From T22, blocks split at the stops are the main scenario; the whole road is the comparison.

Yantai · building 014

In T19 one door was mapped; the review added the second, as on the imagery. On 9 Oct: vehicles can pass at the doors, so holding a door while loading is conservative. → T25 adds door granularity.



On both yards the binding element sat on a conservative modelling choice, and the check changed the model. Three elements per yard, no precise annotation: a minimal human check, written into the thesis as an engineering method.

Fifteen yards: 2 done, 6 in group A next, 7 in group B

■ Done
 ■ Group A: next, reaching eight yards in six countries
 ■ Group B: up to 15

Hanwha Ocean Okpo KR LNG carriers, container ships, offshore Flag-experiment yard; dock-road type. Also builds submarines and naval ships, disclosed	CIMC Raffles Yantai CN rigs, FPSOs, wind installation vessels Offshore yard; door type. Public environmental and cleaner-production filings	Chantiers de l'Atlantique FR cruise ships Large European cruise yard; IGN open imagery, the richest OSM	Meyer Werft Papenburg DE cruise ships Covered building docks, a distinct dock type; LGLN open imagery	Imabari Saijo JP bulk and ore carriers, container ships A building dock over 400 m with three large gantry cranes; GSI imagery
Philly Shipyard US Jones Act tankers, container ships Small single-dock yard; NAIP public-domain imagery	New Yangzi Shipbuilding CN container ships, bulk carriers, tankers Large multi-dock yard; reports of a Singapore-listed group	Samsung Heavy Geoje KR LNG carriers, container ships, offshore On the same island as Okpo; DART filings	Fincantieri Monfalcone IT cruise ships Second European cruise yard, civilian only; regional open imagery	Oshima Shipbuilding JP bulk carriers Mid-size Japanese bulk-carrier yard; GSI imagery
Namura Imari JP bulk carriers, tankers A third Japanese yard; GSI imagery	CM Jinling Weihai CN ro-pax ferries Small yard with a distinct product	Hengli Heavy Industry CN bulk carriers, tankers, container ships Expanded in 2024; imagery must postdate it	HD Hyundai Samho KR VLCCs, container ships, LNG carriers Large yard; DART filings	HD Hyundai Mipo KR MR tankers, mid-size container ships Compact urban yard with four dry docks

Selection looks for spread: dock type, key-element type and the level of daily tasks ÷ T1' matter more than the count.

Source: yard list (project control, 9 Oct 2026), §2; list and reasons checked against public sources, and the entries named at the end of its §6 are verified together with facility use before inclusion. Country codes: KR Korea, CN China, FR France, DE Germany, JP Japan, US United States, IT Italy.

Selection rules: civilian yards, open data first, wide variety

- 1 **Civilian yards:** naval builders are left out; Okpo, the finished flag yard, stays, with its naval business disclosed
- 2 **Open data first:** open-licence orthophotos; OSM roads as tracing drafts; annual reports and environmental filings to check facility use and output
- 3 **Wide variety:** size, layout, dock type (open docks, covered halls, single, multiple), region, product
- 4 **Checks before inclusion:** imagery from 2022 or later; blocks reach the dock by road; after a recent expansion, imagery that postdates it
- 5 **Choose by key-element type and by daily tasks ÷ T1':** spread matters more than count

Group C reserves, for any A or B yard that fails a check

- New Times (Jiangsu) CN
- Shanghai Waigaoqiao CN
- JMU Ariake JP
- Tsuneishi Zhoushan CN
- Daehan Shipbuilding KR
- K Shipbuilding KR

Removed from the list

- **Hudong-Zhonghua, GSI Nansha:** naval shipbuilding
- **Meyer Turku (Finland):** some blocks come by barge from Germany, not by yard road

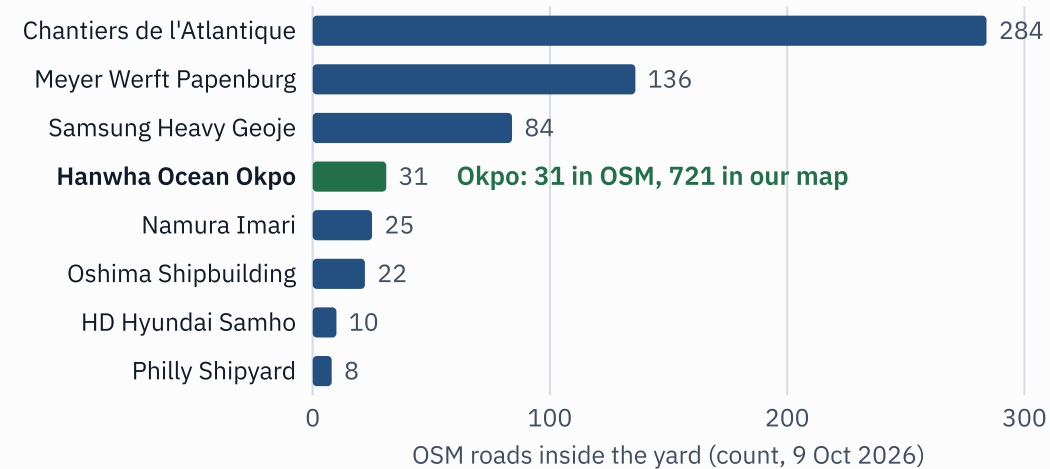
Two cautions

- **Philly Shipyard:** Hanwha plans expansion and naval orders; civilian for now, with dated pre-expansion imagery; swapped for group C if it turns mainly naval
- **Covered dock halls of cruise yards:** if blocks do not reach the dock by road, rule 4 removes the yard

Open imagery and OSM drafts; figures show only our traced vectors

REGION	OPEN IMAGERY	LICENCE
France	IGN BD ORTHO (about 20 cm)	Licence Ouverte 2.0
Lower Saxony, Germany	LGLN DOP20 (20 cm)	CC BY 4.0 (credit © GeoBasis-DE/LGLN)
Friuli Venezia Giulia, Italy	IRDAT / Eagle.fvg orthophotos	IODL 2.0 or CC BY 4.0
Japan	GSI aerial photo tiles	use with attribution
United States	USDA NAIP (about 60 cm)	public domain
China, Korea	no openly licensed high-resolution imagery	trace vectors only; no imagery in figures

Licence terms and capture years are checked against the originals on inclusion (IGN, NAIP and GSI terms to be checked); base-map dates are to be checked except Philly (NAIP, 10 May 2022). Chinese and Korean yards have rich public filings, used to verify facility use and output.

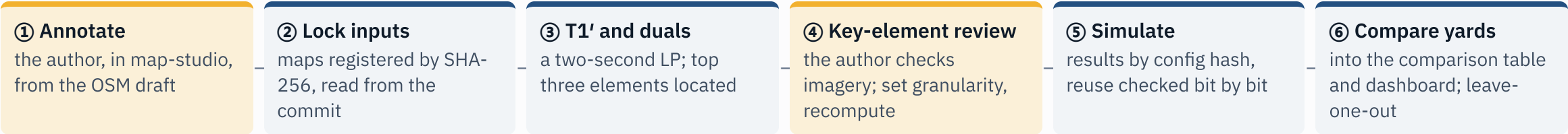


- **OSM drafts for 7 yards have base maps** (5 open imagery, the 2 Korean ones cut from existing imagery); in map-studio the author checks, adds roads, measures widths, marks facilities and entrances
- **Data sharing:** OSM-derived parts are shared alike under ODbL
- **Optional accelerator:** MY-A's segmentation extracts roads, yards and buildings from imagery where OSM is sparse

[Open the segmentation \(MY-A\) briefing →](#)

Sources: yard list, imagery table of §1 and §2 (OSM road counts from Overpass, 9 Oct 2026, © OpenStreetMap contributors, ODbL); tasks.md, multi-yard annotation (9 Oct 2026, drafts and base maps); review §2.7 item 4 (imagery terms, ODbL). No imagery or network geometry is shown.

Per yard: bound first, then the key-element review, then simulation



STEP	WHO	STATUS	OUTPUT (CHAPTER 6)
Yard list; OSM drafts and base maps for 7 yards	curation	done	Table of yards and data sources
Annotating group A's 6 yards	author	running	Multi-yard instance library (vector networks)
T1' and top-3 duals per yard; two OD samplings	computation	on each map	Comparison table: T1', binding element, type
Key-element review, then recompute	author	fixed step	Review record (map source field)
Verify facility use and output; one folding method for daily tasks	curation, library	per yard	Density chart: daily tasks ÷ T1'
Flag simulations with T22's rules and task volumes	computation	after T1'	Plateau, K*, capacity interval
Topology of 8 yards; order group B by the gaps	curation	after A	Indicator table that explains
Leave-one-out and types: a round at 8, final at 15	—	planned	Main figure: prediction against simulation; type table

Sources: yard list §3; tasks.md, multi-yard annotation; review §1.4 (inputs locked by SHA-256, results by config hash, reuse checked bit by bit) and §4.3–4.4 (chapter 6 plan; weeks 5–8 compute T1' and plateaus for the first yards of group A).

What is established, and what is in progress

Established (thesis-core 83c16bf)

- T1' runs on both yards: none of 17,760 occupancy points exceeds it
- Both yards have a plateau independent of fleet size; K^* is 50 on Okpo and 25 on Yantai
- Two key-element types: dock-road single-lane and door
- In the main mix both reservation plateaus are about 36% of T1'
- The key-element review changed the model on both yards and is now a fixed step
- The pipeline: yard list, drafts and base maps for 7 yards, SHA-256-locked inputs, results by config hash, bit-by-bit reuse checks

TASK	CONTENT	WHAT IT BRINGS TO CHAPTER 6
T22 running	Safe release rule segment_safe (Banker-type deadlock avoidance) and a no-clearing switch; dock road in blocks with direction locks; whole-yard task volumes (300 / 450 / 600, peak $\times 2$, 16 h and 24 h)	Rules and task-volume caliber for every yard; Okpo's whole-yard density reading
T23 running	System T1' without sampling noise; route-free T1'' and the value of route choice; top-10 dual prices	The predictor itself; the list of binding elements
T24 running	External anchors and citation checks; inputs for Okpo's daily task volume	One folding method across yards
T25 queued	Upgrade simulations and the first road-vehicle exchange-rate table; door granularity; unambiguous resource names	The upgrade-value target; Yantai's door granularity
T26 queued	Okpo on rev 1833 (floating docks, branch roads); floating-dock scenario; finite holding area S and $K_0(S)$	Okpo's final layout

Sources: thesis-core 83c16bf (T17–T21); the 17,760 points in the T21 report, §0 item 6; task contents from the T22–T26 rows of tasks.md (9 Oct 2026). Running and queued tasks are plans, not results.

Risks and mitigations

RISK	MITIGATION
Plateau ÷ T1' moves with the traffic rule, the OD sampling and how the key element is modelled (67% for Yantai with one door)	Read per rule and per type; two OD samplings as each yard's standard sensitivity; review the key element before reading the ratio
The binding element sits on a conservative modelling choice (on both yards)	The key-element review is a fixed step; every reading states its resource granularity
Fifteen samples are few	No regression: T1' as the predictor, grouping by key element, leave-one-out; spread matters more than count
Daily task volumes come in different calibers	One folding method (steel ÷ block weight × moves ÷ working days) checked against public filings; Okpo's own caliber from T22 and T24
Imagery licences and capture dates not yet checked one by one	Check the originals on inclusion; imagery from 2022 on and after expansions; figures show vectors only; OSM-derived parts under ODbL
A yard fails after selection (turns mainly naval, or blocks arrive by sea rather than by road)	Six group C reserves step in; Philly and covered dock halls already have rules

Sources: review and follow-up guidance §2.6, §2.7, §5 and §3 question 6; yard list §1–2; T19 report §4; T20's r383 → r395 comparison (thesis-core 83c16bf).

Candidate journals (chosen before submission) and a working title

Working title (direction)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts

The question of the direction document, reframed around H6 of the research manifesto: the bound and its binding element are the predictors

FIRST CHOICE · SHIP AND INDUSTRIAL
ENGINEERING

Ocean Engineering

Readers of engineering methods for shipyard production and logistics; named for chapter 6 in the direction document

Computers & Industrial Engineering

Simulation and capacity analysis in industrial engineering; named in the direction document

ALTERNATIVES · PRODUCTION AND TRANSPORT

International Journal of Production Research

Production logistics and scheduling; where Roh & Cha 2011 appeared

Transportation Research Part E

Capacity and design of logistics and transport systems

APPLIED · MARITIME

Journal of Marine Science and Engineering

Marine engineering, open access; where Wang et al. 2024 published their yard-delay survey

Maritime Policy & Management

Management readers in shipping, ports and yards

Three parts: the map-computed bound and binding element as cross-yard predictor (H6); key-element review and open-data pipeline (engineering method); vector networks of 15 public layouts (OSM parts under ODbL).

References

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Data and material: thesis-core commit 83c16bf (results and reports of T17–T21) and tag manifesto-v1 (research manifesto v1); the project's yard list and the review and follow-up guidance (both 9 Oct 2026); OpenStreetMap contributors, queried through the Overpass API (9 Oct 2026), ODbL 1.0; open imagery IGN BD ORTHO, LGLN DOP20, IRDAT FVG / Eagle.fvg, GSI tiles, USDA NAIP (terms checked on inclusion); CIMC Raffles cleaner-production notice (29 Apr 2024) and environmental disclosure report for 2021.

Back to the thesis briefing, or on to the neighbouring studies

The predictor is T1' itself

One two-second linear program per yard; both yards' reservation plateaus sit at about 36% of T1', each predicting the other within 0.5%.

Key-element review is a fixed step

After T1', the author checks the top-three dual-price elements on imagery and then sets the resource granularity; in both yards the check changed the model.

Fifteen public layouts

Two done, six group-A yards being traced; grouped by key-element type and tested leave-one-out on plateau, K^* and upgrade value.

Bound first, binding element next, simulation last: chapter 6 tests whether a two-second linear program can screen a new shipyard.

[← Back to the thesis briefing \(road-occupying transport\)](#)[Occupancy model and bounds →](#)[Capacity interval →](#)[Fleet-network co-design →](#)[The fleet-sizing \(MY-B\) briefing →](#)[The segmentation \(MY-A\) briefing →](#)

Thank you.